

Multinational Firms and Agricultural Trade in Latin America

Ownership and Exports in Agriculture

Ignacio Marra de Artiñano* Gabriel Scattolo[†] Sebastián Velásquez[‡]
Christian Volpe Martincus[§]

This version: September 2026

Preliminary draft, please do not circulate

Abstract

We match the customs export records of nine Latin American countries to the ownership records of Orbis and Dun & Bradstreet and describe who owns the region's agricultural exporters. Multinational affiliates account for 48 percent of agricultural exports (HS 01–24), 38 percent foreign owned and 10 percent domestically owned. The share ranges from 29 percent in El Salvador to 66 percent in Paraguay. The parents come from a few countries, and a small number of large groups accounts for half of foreign-affiliate exports. Unlike in the region's exports as a whole, foreign affiliates in agriculture are not concentrated in more complex or more differentiated products. Product–destination pairs with more multinationals export more, and distance reduces the exports of multinationals less than those of other firms.

Keywords: Multinational Firms, Agricultural Trade, Ownership, Firm-Level Data, Latin America

JEL codes: F14, F23, Q17, L25

*Université Libre de Bruxelles (ECARES).

[†]Inter-American Development Bank.

[‡]Inter-American Development Bank.

[§]Inter-American Development Bank and CESifo. The views and interpretations in this paper are strictly those of the authors and should not be attributed to the Inter-American Development Bank, its executive directors, or its member countries.

Contents

1	Introduction	3
2	Related Literature	3
3	Data	3
3.1	Customs data	3
3.2	Ownership data and matching	3
3.3	Definitions	4
3.4	Sample	4
4	Stylized Facts in Agriculture	5
4.1	Fact 1: Multinationals Account for a Large Share of Agricultural Exports	5
4.2	Fact 2: Product Sophistication Does Not Sort Owners in Agriculture	6
4.3	Fact 3: Multinationals from a Small Set of Countries Dominate Exports	8
4.4	Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports	9
4.5	Fact 5: Multinational Presence Is Associated with Higher Exports	10
4.6	Fact 6: Distance Is a Weaker Barrier for Multinationals	11
5	Product Groups within Agriculture	12
6	Theoretical Framework	14
7	Concluding Remarks	15
	References	15
A	Additional Tables and Figures	16

1 Introduction

Customs records show what a country exports and where, but not who owns the exporter. In agriculture this matters because a large part of Latin America’s exports is handled by traders, processors and food companies headquartered abroad. Foreign direct investment statistics show where capital goes, not which firms export which products to which markets.

In this paper we match the customs export records of nine countries (Argentina, Chile, Colombia, Costa Rica, the Dominican Republic, El Salvador, Paraguay, Peru and Uruguay) to the ownership records of Orbis and Dun & Bradstreet, classify every exporter as a foreign multinational affiliate, a domestic multinational affiliate or a local firm, and focus on agricultural exports (HS chapters 01–24). These amount to US\$78.9 billion a year, a third of the nine countries’ exports. All results are for the nine countries together.

Multinational affiliates account for 48 percent of agricultural exports, 38 percent foreign owned and 10 percent domestically owned, with shares that range from 29 percent in El Salvador to 66 percent in Paraguay. Parents come from a few countries, and groups with more than 100 affiliates worldwide, 12 percent of all parents, account for half of foreign-affiliate exports. Among the five largest product groups, the foreign share is highest in oil seeds (62 percent) and cereals (49 percent). Two patterns found in the region’s exports as a whole do not hold in agriculture: foreign affiliates are not concentrated in more complex products or in differentiated products. The other results do hold. Product–destination pairs with more multinationals export more, and distance reduces the exports of multinationals less than those of other firms.

Contribution to the literature: to be written.

Section 2 reviews the literature, Section 3 describes the data and Section 4 presents the facts. Section 5 looks at product groups within agriculture, Section 6 outlines the model and Section 7 concludes.

2 Related Literature

To be written. Three strands: firm-level evidence on multinational production and trade; multinationals in agricultural and food trade; and evidence on multinational firms in Latin America.

3 Data

3.1 Customs data

The export records come from the customs authorities of the nine countries. Each record gives the value exported (FOB, US dollars) by a firm, of an HS6 product (2007 revision), to a destination, in a year. Firms are identified by their tax number and name. The records cover all merchandise exports in the years available for each country: Argentina 2011–2019, Chile 2009–2022, Colombia 2010–2021, Costa Rica 2010–2019, the Dominican Republic 2012–2019, El Salvador 2006–2018, Paraguay 2012–2020, Peru 2010–2019 and Uruguay 2010–2019.

Agricultural exports are HS chapters 01–24: live animals, meat, fish, dairy, vegetables, fruit, cereals, oil seeds, fats and oils, prepared foods, beverages and tobacco. All results in this paper are for these products and for the nine countries together.

3.2 Ownership data and matching

Ownership comes from Orbis (Bureau van Dijk) and Dun & Bradstreet, which report for each company its global ultimate owner and the owner’s country. We combine the two sources into one file with the group, the country of the parent and the number of affiliates the group has worldwide.

Exporters are matched to this file in four steps: by tax number where the corporate records report it; by company name within each country, using a character-based similarity measure, with each candidate pair reviewed by a large language model; by company name across all countries; and, for the 500 largest unmatched exporters of each country, by hand, using company websites and business registries. The tax number delivers about 60 percent of matched export value. After matching, the parent’s country is known for 92.5 percent of the export value of foreign affiliates.

3.3 Definitions

An exporter found in Orbis or Dun & Bradstreet is a multinational affiliate (MNE). It is a *domestic* MNE when its ultimate parent is located in the exporting country and a *foreign* MNE otherwise; matched exporters without a recorded parent country are counted as foreign. Exporters not found in either database are *local* firms. Because the match relies on ownership links, a group’s head company exporting from its own country cannot be identified and appears among local firms.

Products are classified with the Product Complexity Index of Hausmann and Hidalgo, the technology classes of Lall (2000), the Rauch (1999) classification, and the end-use classes of BEC Rev. 4. Sectors are defined on HS chapters: agriculture 01–24, mining and fuels 25–27 and 71, manufacturing 28–97 excluding 71. Distance comes from the CEPII gravity database.

3.4 Sample

Table 1 describes the agricultural sample by country and Table 2 summarizes it. The nine countries have 38,832 agricultural exporters, of which 7,993 are matched to a corporate record, and export 1,023 HS6 products to 245 destinations.

Table 1: Agricultural Sample Coverage, by Exporting Country

Origin	Years	Exporters	Matched	HS6 lines	Destinations	Records	Value (US\$bn/yr)
ARG	2011–2019	6,709	1,940	686	195	217,027	38.1
PER	2010–2019	7,366	1,219	658	193	163,296	8.1
CHL	2009–2022	5,015	1,833	783	196	245,797	7.5
COL	2010–2021	7,816	1,458	718	206	164,576	7.2
PRY	2012–2020	822	252	272	155	16,893	6.2
URY	2010–2019	1,269	355	510	188	34,132	4.4
CRI	2010–2019	3,776	362	700	174	80,079	4.3
DOM	2012–2019	4,677	434	601	172	95,340	2.0
SLV	2006–2018	1,382	140	610	125	33,701	1.1
All nine	2006–2022	38,832	7,993	1,023	245	1,050,841	78.9

Notes: Agricultural exports: HS 01–24. Exporters are distinct tax identifiers within each country; matched exporters are those found in Orbis or Dun & Bradstreet. Records are firm–destination–HS6–year observations. Values are annual averages in US\$ billion.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 2: Agricultural Sample Summary

Statistic	Value
Exporting countries	9
Years	2006–2022
Records (firm $\times$ destination $\times$ HS6 $\times$ year)	1,050,841
Exporters	38,832
matched to Orbis / Dun & Bradstreet	7,993
HS6 product lines	1,023
Destination markets	245
Parent countries of foreign multinationals	130
Multinational groups	4,584
Export value, all years (US\$ billion)	788
foreign multinationals	38.1%
domestic multinationals	9.5%
local firms	52.4%
Foreign-MNE value with a recorded parent country	87.7%

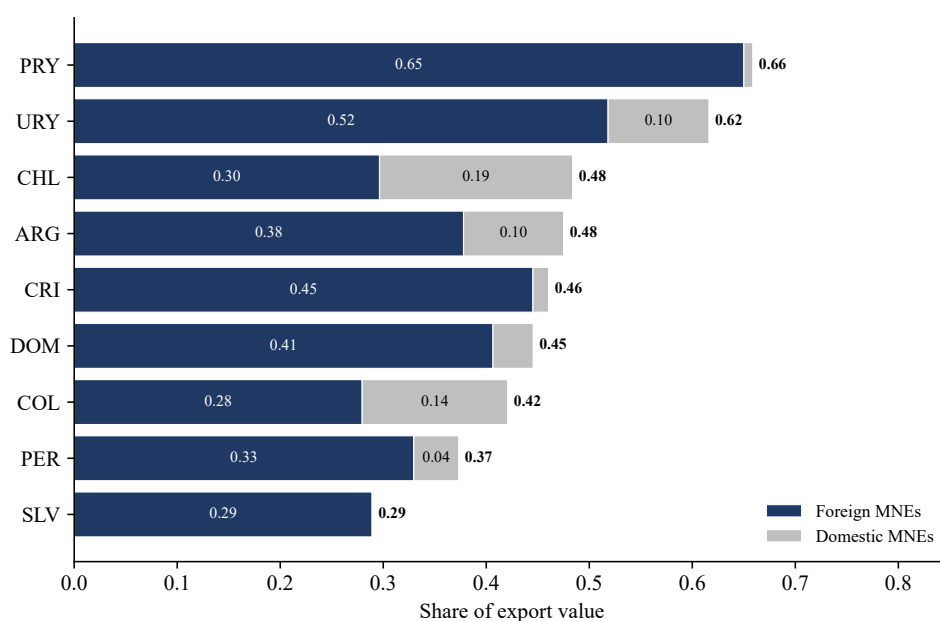
Notes: Agricultural exports of the nine countries, all available years pooled. Shares are percent of agricultural export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4 Stylized Facts in Agriculture

4.1 Fact 1: Multinationals Account for a Large Share of Agricultural Exports

Figure 1 shows the share of each country's agricultural exports accounted for by multinational affiliates. It is 48 percent for the nine countries together, 38 percent foreign owned and 10 percent domestically owned. It ranges from 66 percent in Paraguay and 62 percent in Uruguay to 37 percent in Peru and 29 percent in El Salvador. Foreign affiliates export more than domestic multinationals in every country; the gap is smallest in Chile (30 against 19 percent).

**Figure 1: Multinational Share of Agricultural Export Value, by Exporting Country**

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. The figure at the end of each bar is the total MNE share.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.2 Fact 2: Product Sophistication Does Not Sort Owners in Agriculture

In the region's exports as a whole, the foreign share rises with product complexity and is higher for differentiated products. Neither pattern holds in agriculture. Across quintiles of the Product Complexity Index the foreign share is 0.29, 0.49, 0.38, 0.42 and 0.55 (Figure 2). By Rauch class it is 41 percent for differentiated products, 32 percent for reference-priced products and 41 percent for homogeneous products, and the domestic share is about 9 percent in all three (Table 3 and Figure 3). By end use, the foreign share is 41 percent for intermediate goods and 35 percent for consumption goods (Table 4).

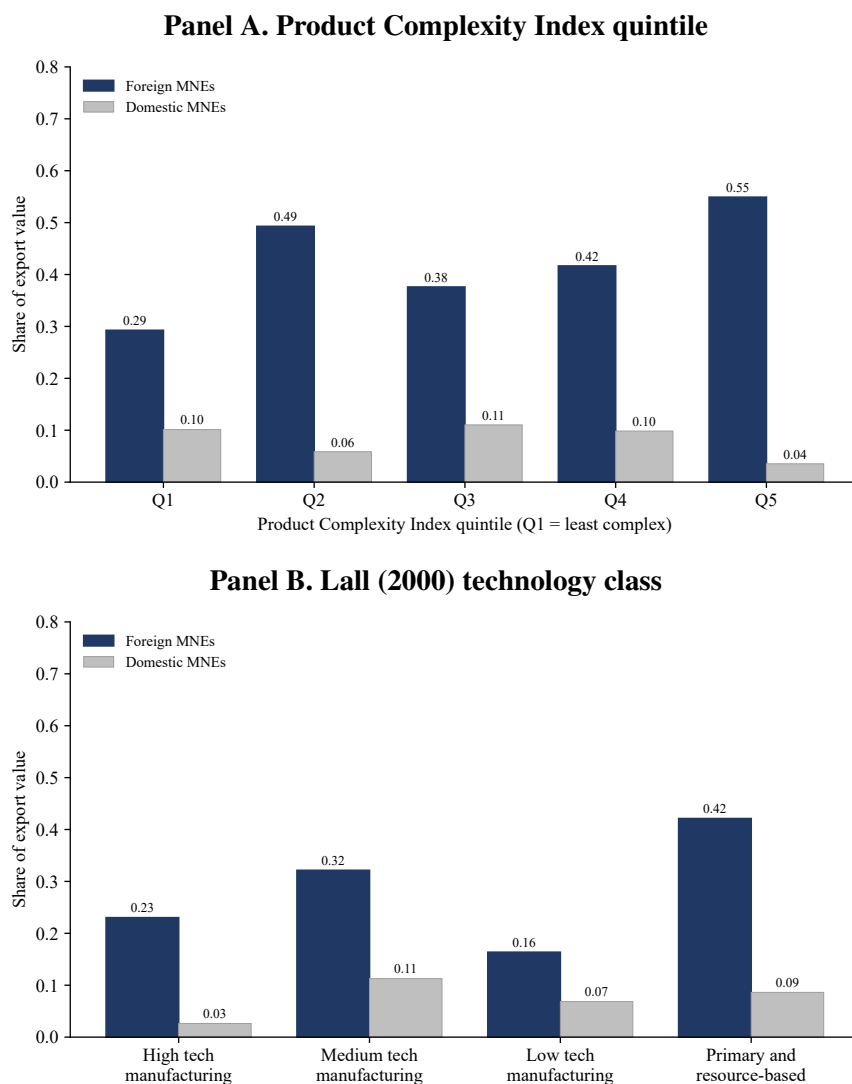


Figure 2: Multinational Share of Agricultural Export Value, by Product Sophistication

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

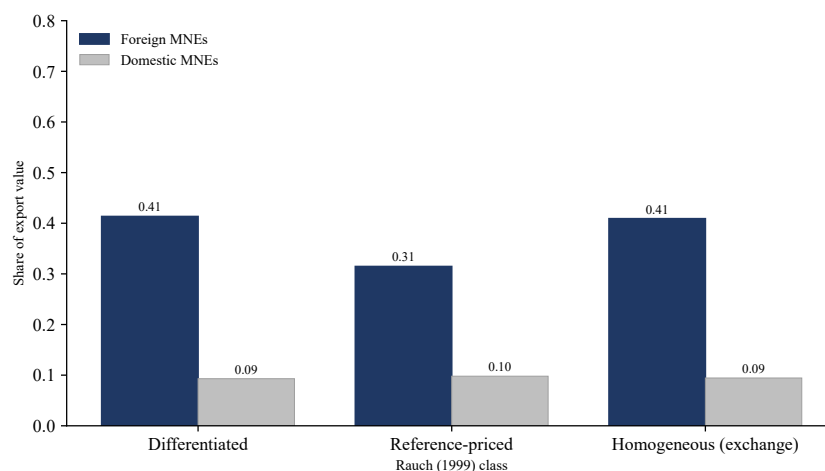
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 3: Multinational Share of Agricultural Export Value, by Rauch Product Class

Rauch (1999) class	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Differentiated	0.413	0.093	0.506	167	7.8
Reference-priced	0.315	0.098	0.413	357	20.1
Homogeneous (exchange)	0.409	0.094	0.503	164	43.7

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 3: Multinational Share of Agricultural Export Value, by Rauch Product Class**

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 4: Multinational Share of Agricultural Export Value, by End Use

BEC end use	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Intermediate	0.411	0.090	0.501	243	42.5
Consumption	0.351	0.099	0.450	456	30.4
Capital	0.168	0.020	0.187	3	0.0

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. End use from BEC Rev.4; HS6 products without a BEC correspondence are excluded. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 5: Product Sophistication and Multinational Shares in Agriculture

	PCI		Lall		Rauch	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: share of all MNEs</i>						
Complexity (PCI)	0.0733*** (0.0061)	0.0614*** (0.0064)				
Lall: high tech (vs primary and resource-based)			-0.2016*** (0.0104)	-0.2103*** (0.0102)		
Lall: medium tech			-0.0395*** (0.0060)	-0.0465*** (0.0066)		
Lall: low tech			-0.2467*** (0.0278)	-0.2272*** (0.0268)		
Rauch: differentiated (vs homogeneous)					0.0435*** (0.0088)	0.0488*** (0.0098)
Rauch: reference-priced					-0.0460*** (0.0068)	-0.0393*** (0.0066)
Observations	158,616	156,478	274,948	272,985	271,105	269,190
<i>Panel B. Dependent variable: share of foreign MNEs</i>						
Complexity (PCI)	0.0562*** (0.0067)	0.0490*** (0.0072)				
Lall: high tech (vs primary and resource-based)			-0.1480*** (0.0091)	-0.1576*** (0.0094)		
Lall: medium tech			-0.0623*** (0.0067)	-0.0678*** (0.0073)		
Lall: low tech			-0.2301*** (0.0153)	-0.2262*** (0.0139)		
Rauch: differentiated (vs homogeneous)					0.0554*** (0.0097)	0.0560*** (0.0107)
Rauch: reference-priced					-0.0439*** (0.0069)	-0.0383*** (0.0066)
Observations	158,616	156,478	274,948	272,985	271,105	269,190
<i>Panel C. Dependent variable: share of domestic MNEs</i>						
Complexity (PCI)	0.0172*** (0.0031)	0.0123*** (0.0033)				
Lall: high tech (vs primary and resource-based)			-0.0536*** (0.0051)	-0.0527*** (0.0053)		
Lall: medium tech			0.0229*** (0.0032)	0.0212*** (0.0033)		
Lall: low tech			-0.0165 (0.0247)	-0.0010 (0.0273)		
Rauch: differentiated (vs homogeneous)					-0.0120*** (0.0044)	-0.0071* (0.0043)
Rauch: reference-priced					-0.0021 (0.0038)	-0.0010 (0.0040)
Observations	158,616	156,478	274,948	272,985	271,105	269,190
Origin × year and destination × year FE	✓		✓		✓	
Origin × dest. × year FE		✓		✓		✓

Notes: Observations are origin–destination–HS6–year cells. Agricultural products only. Dependent variable: share of the cell’s export value shipped by each type of MNE. Omitted categories: primary and resource-based products (Lall) and homogeneous products (Rauch). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.3 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

Figure 4 ranks parent countries by their share of foreign-affiliate agricultural exports. The ten largest parent countries account for more than 70 percent. The list reflects the grain trade and the jurisdictions where trading groups hold their assets: Switzerland (11 percent), the Netherlands (9.5 percent) and Liechtenstein (7.6 percent) rank high, together with the United States (24 percent). The destination of each parent country’s exports is in Tables A.5 and A.6 in the appendix.

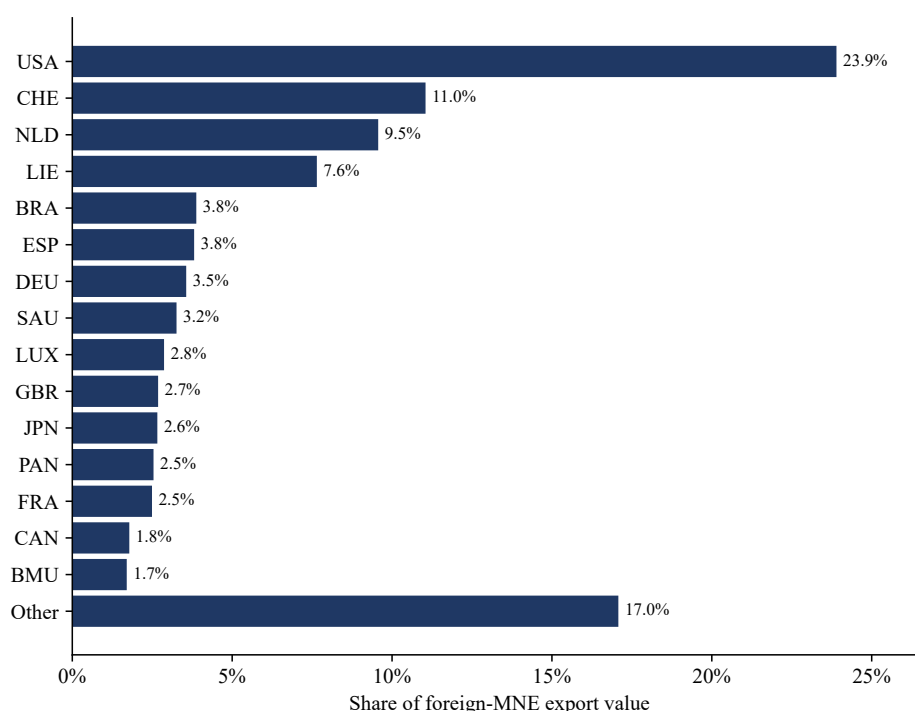


Figure 4: Foreign-Multinational Agricultural Exports, by Parent Country

Notes: Percent of foreign-MNE agricultural export value, by country of the ultimate parent.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.4 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

Figure 5 groups parents by the number of affiliates their group has worldwide. In agriculture 45 percent of parents have a single affiliate, but the 12 percent with more than 100 affiliates account for half of foreign-affiliate exports.

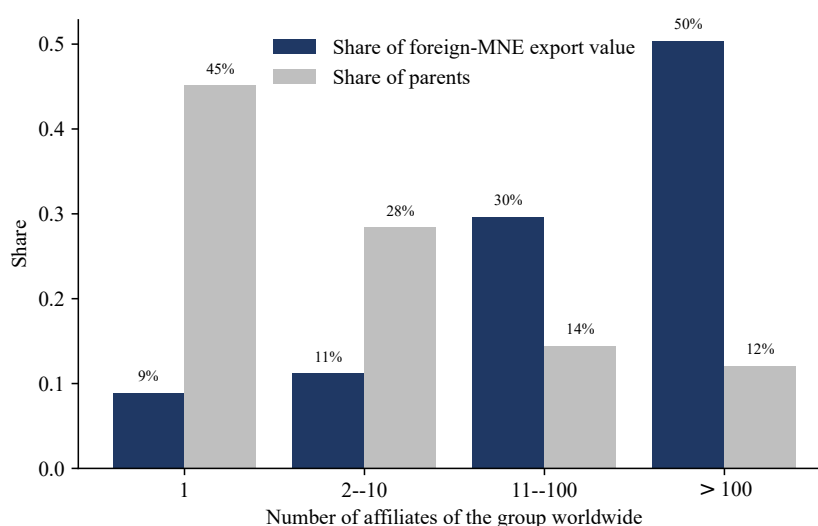


Figure 5: Foreign-Multinational Agricultural Exports and Number of Parents, by Size of the Group's Global Network

Notes: Shares of foreign-MNE agricultural export value and of parents, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.5 Fact 5: Multinational Presence Is Associated with Higher Exports

Table 6 relates the agricultural exports of an origin–destination–product–year cell to the number of multinationals exporting in it and to whether any does. The results are similar to those for all exports, including for the exports of local firms in the same cell (Panel B). Table 7 splits the intensive margin by type of multinational: the coefficient is 1.30 for foreign affiliates and 0.52 for domestic ones.

Table 6: Multinational Presence and Agricultural Exports

	Intensive: ln(# MNE firms)			Extensive: any MNE present		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.1821*** (0.0283)	2.1861*** (0.0279)	0.8509*** (0.0190)			
Any MNE present				1.7857*** (0.0611)	1.7863*** (0.0604)	0.6860*** (0.0232)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.5538*** (0.0332)	1.5628*** (0.0332)	0.4048*** (0.0245)			
Any MNE present				1.9365*** (0.0567)	1.9313*** (0.0554)	0.2812*** (0.0181)
Origin FE	✓			✓		
Destination FE	✓			✓		
Product FE	✓	✓		✓	✓	
Year FE	✓			✓		
Origin × year FE		✓			✓	
Destination × year FE		✓			✓	
Origin × dest. × product FE			✓			✓
Origin × dest. × year FE			✓			✓
Observations (Panel A)	154,199	153,847	137,324	296,092	295,788	268,332
Observations (Panel B)	91,020	90,655	81,670	232,907	232,617	208,479

Notes: Observations are origin–destination–HS6–year cells. Agricultural products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 7: Multinational Presence and Agricultural Exports: Intensive Margin, by Type of Multinational

	All MNEs		Foreign / domestic		Foreign by destination, domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.1861*** (0.0279)	0.8509*** (0.0190)				
ln(# foreign MNEs)			1.2954*** (0.0275)	0.4625*** (0.0188)		
ln(# foreign MNEs, parent-country destination)					0.4928*** (0.1145)	0.2334*** (0.0804)
ln(# foreign MNEs, other destinations)					0.9569*** (0.0754)	0.3804*** (0.0360)
ln(# domestic MNEs)			0.5233*** (0.0266)	0.2831*** (0.0234)	0.4096*** (0.0753)	0.1935*** (0.0421)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.5628*** (0.0332)	0.4048*** (0.0245)				
ln(# foreign MNEs)			1.2191*** (0.0363)	0.3602*** (0.0292)		
ln(# foreign MNEs, parent-country destination)					0.3864*** (0.1357)	0.2698** (0.1255)
ln(# foreign MNEs, other destinations)					1.1003*** (0.1197)	0.4395*** (0.0693)
ln(# domestic MNEs)			0.5676*** (0.0339)	0.2461*** (0.0341)	0.4367*** (0.1236)	0.2032** (0.0759)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (Panel A)	153,847	137,324	25,437	22,002	1,952	1,604
Observations (Panel B)	90,655	81,670	22,503	19,535	1,903	1,567

Notes: Observations are origin–destination–HS6–year cells. Agricultural products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.6 Fact 6: Distance Is a Weaker Barrier for Multinationals

Table 8 estimates how firm agricultural exports fall with distance. The elasticity is -0.21 for local firms, larger in absolute value than for all exports. Table 9 shows that it is 0.037 smaller in absolute value for foreign affiliates and 0.019 smaller for domestic ones.

Table 8: Distance and Firm Agricultural Exports: Multinationals by Presence at the Destination

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.2118*** (0.0748)	-0.2269*** (0.0712)	-0.2078*** (0.0745)	-0.2224*** (0.0708)
ln distance × MNE	0.0326*** (0.0031)	0.0318*** (0.0031)		
ln distance × MNE, present at destination			0.0890*** (0.0070)	0.0875*** (0.0069)
ln distance × MNE, not present at destination			0.0228*** (0.0032)	0.0221*** (0.0032)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	1,045,009	1,044,787	1,045,009	1,044,787

Notes: Observations are firm–destination–HS6–year cells. Agricultural products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 9: Distance and Firm Agricultural Exports: Foreign and Domestic Multinationals

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.2135*** (0.0748)	-0.2285*** (0.0711)	-0.2130*** (0.0748)	-0.2280*** (0.0711)
ln distance × foreign MNE	0.0366*** (0.0034)	0.0358*** (0.0034)		
ln distance × foreign MNE, parent-country destination			0.0610*** (0.0165)	0.0590*** (0.0165)
ln distance × foreign MNE, other destination			0.0359*** (0.0035)	0.0351*** (0.0034)
ln distance × domestic MNE	0.0187*** (0.0046)	0.0182*** (0.0046)	0.0187*** (0.0046)	0.0182*** (0.0046)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	1,045,009	1,044,787	1,045,009	1,044,787

Notes: Observations are firm–destination–HS6–year cells. Agricultural products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

5 Product Groups within Agriculture

Table 10 and Figure 6 split agricultural exports by SITC division. The largest divisions are animal feed (US\$13.8 billion a year), vegetables and fruit (10.6), cereals (10.3), oil seeds (6.7) and vegetable oils (6.0). The foreign share is highest in miscellaneous edible products (71 percent), oil seeds (62 percent) and tobacco (55 percent), and lowest in coffee, tea and cocoa (30 percent), beverages (29 percent) and sugar (19 percent). Domestic multinationals are largest in dairy (20 percent), coffee (16 percent) and sugar (15 percent). Figure 7 compares agricultural inputs with other agricultural products, and Figure 8

shows the twenty largest agricultural export products. Other breakdowns are in the appendix.

Table 10: Agricultural Exports and Multinational Shares, by SITC Division

SITC Rev. 3 division	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
08 Animal feed	13.8	6.6	35.7	11.5	CHE 0.28, LIE 0.19, USA 0.18
05 Vegetables & fruit	10.6	4.0	32.6	5.8	ESP 0.14, USA 0.13, PAN 0.11
04 Cereals	10.3	5.9	48.6	8.2	USA 0.33, CHE 0.14, NLD 0.14
22 Oil seeds	6.7	4.4	61.6	4.1	USA 0.54, NLD 0.14, LIE 0.08
07 Coffee, tea, cocoa, spices	5.1	2.3	30.4	15.6	DEU 0.41, ITA 0.14, SGP 0.12
42 Fixed vegetable oils	6.0	2.5	30.8	10.2	CHE 0.27, NLD 0.24, USA 0.23
01 Meat	5.2	2.4	34.8	13.1	SAU 0.46, BRA 0.26, JPN 0.08
03 Fish & seafood	3.4	1.8	41.9	9.5	ESP 0.32, LUX 0.11, ITA 0.10
29 Crude animal/veg. materials	2.2	0.8	27.3	9.0	DEU 0.10, ARG 0.09, BGR 0.08
11 Beverages	2.2	0.8	28.6	9.2	LUX 0.17, PAN 0.15, GBR 0.12
02 Dairy & eggs	1.9	1.0	28.0	20.3	NLD 0.20, CHE 0.19, ROU 0.19
09 Misc. edible products	1.7	1.2	70.9	2.3	USA 0.46, CHE 0.17, FRA 0.09
06 Sugar & honey	1.4	0.5	19.3	14.7	USA 0.50, KOR 0.18, LIE 0.10
12 Tobacco	1.2	0.7	54.8	0.2	USA 0.36, HKG 0.25, GBR 0.12
41 Animal oils & fats	0.5	0.2	23.0	11.0	BMU 0.31, PAN 0.23, GBR 0.16
Other	0.6	0.1	21.6	1.9	CHE 0.21, USA 0.20, ROU 0.10
All	72.9	35.0	38.5	9.4	

Notes: Foreign and domestic: percent of the division's export value. HS6 products without an SITC correspondence (US\$6.0 billion a year) are excluded. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

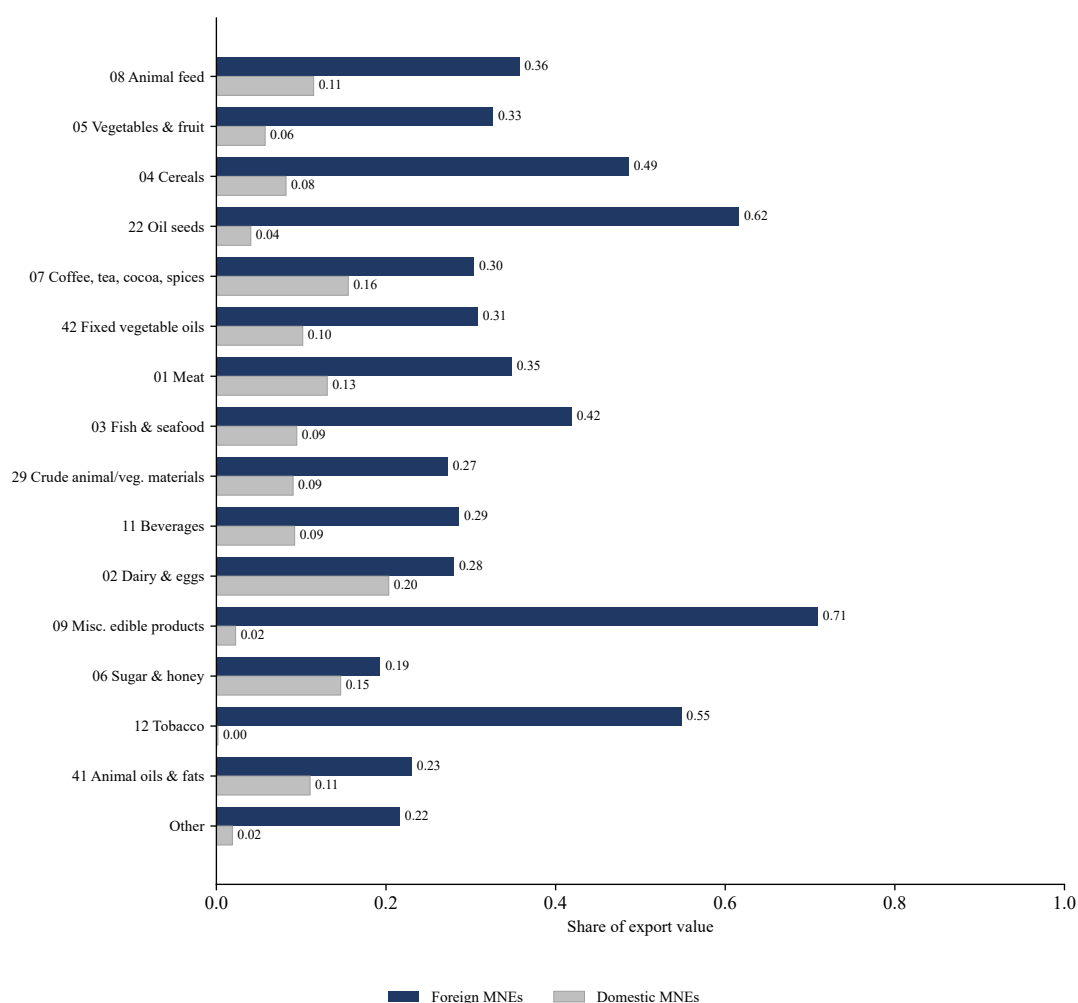


Figure 6: Multinational Share of Agricultural Export Value, by SITC Division

Notes: Shares of each division's export value, all available years pooled. Divisions ordered by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

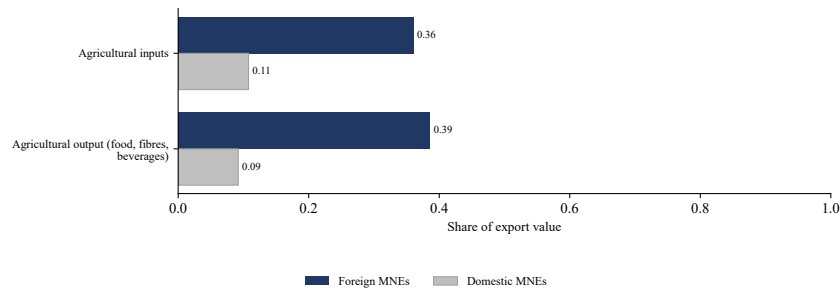


Figure 7: Multinational Share of Agricultural Export Value: Inputs and Other Products

Notes: Shares of export value, all available years pooled. Inputs: seeds, animal feed and live animals.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

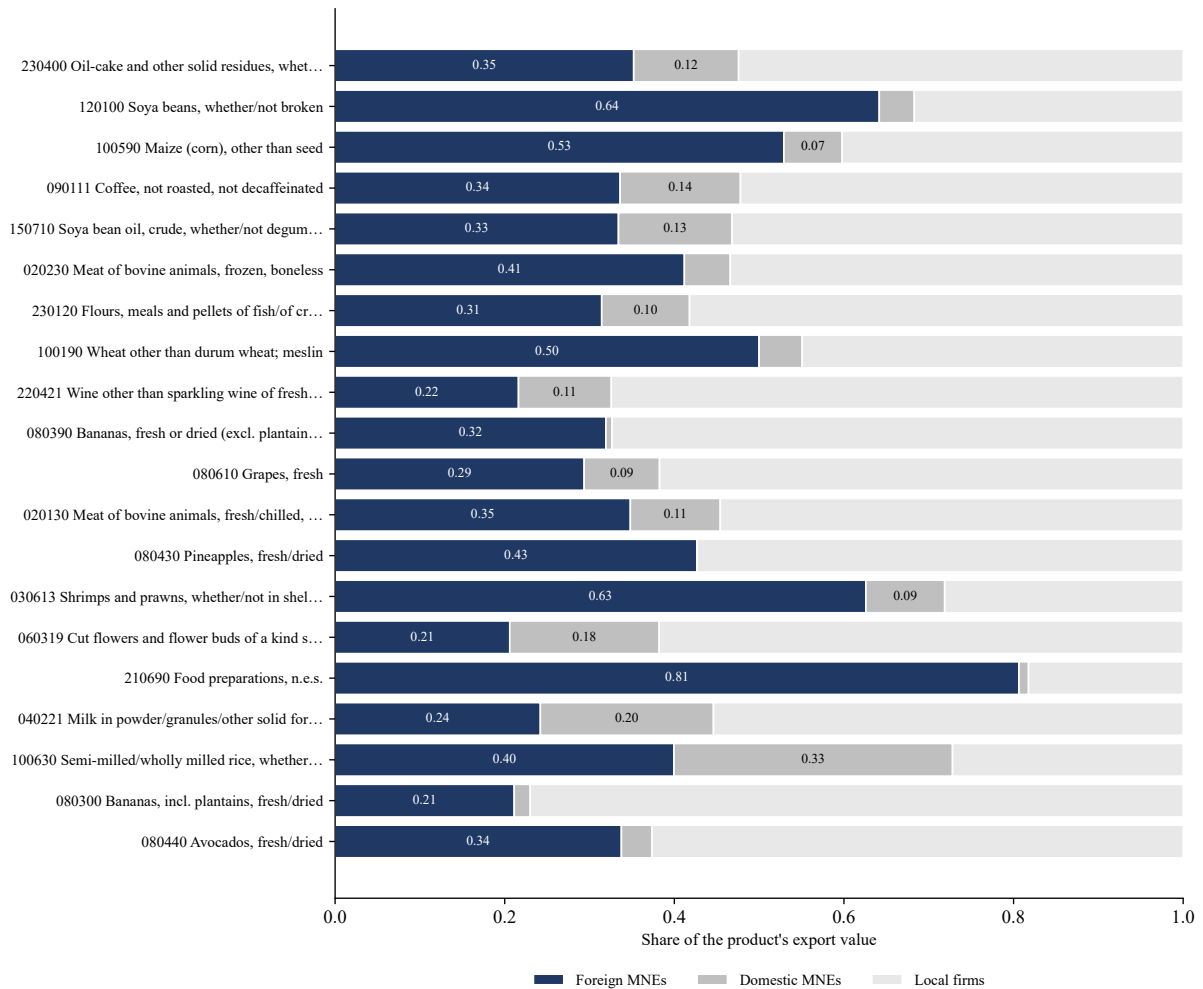


Figure 8: Ownership of the Twenty Largest Agricultural Export Products

Notes: Shares of each product's export value. Twenty largest agricultural HS6 products by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

6 Theoretical Framework

To be written. The model is developed in the companion theory paper.

7 Concluding Remarks

To be written.

References

To be completed.

Hausmann, R. and Hidalgo, C., 2011. The Network Structure of Economic Output. *Journal of Economic Growth*, 16(4).

Lall, S., 2000. The Technological Structure and Performance of Developing Country Manufactured Exports, 1985–98. *Oxford Development Studies*, 28(3).

Rauch, J., 1999. Networks versus Markets in International Trade. *Journal of International Economics*, 48(1).

United Nations Statistics Division, 2022. *Classification by Broad Economic Categories*, Rev. 4. New York.

A Additional Tables and Figures

Fact 1

Table A.1: Multinational Share of Agricultural Export Value, by Exporting Country

Origin	Foreign	Domestic	All MNEs	Value (US\$bn/yr)
PRY	0.650	0.009	0.659	6.2
URY	0.518	0.098	0.616	4.4
CHL	0.296	0.187	0.484	7.5
ARG	0.378	0.097	0.475	38.1
CRI	0.445	0.015	0.461	4.3
DOM	0.407	0.039	0.446	2.0
COL	0.280	0.141	0.421	7.2
PER	0.330	0.044	0.373	8.1
SLV	0.289	0.000	0.289	1.1
All	0.381	0.095	0.476	78.9

Notes: Shares of each country's agricultural export value, all available years pooled. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

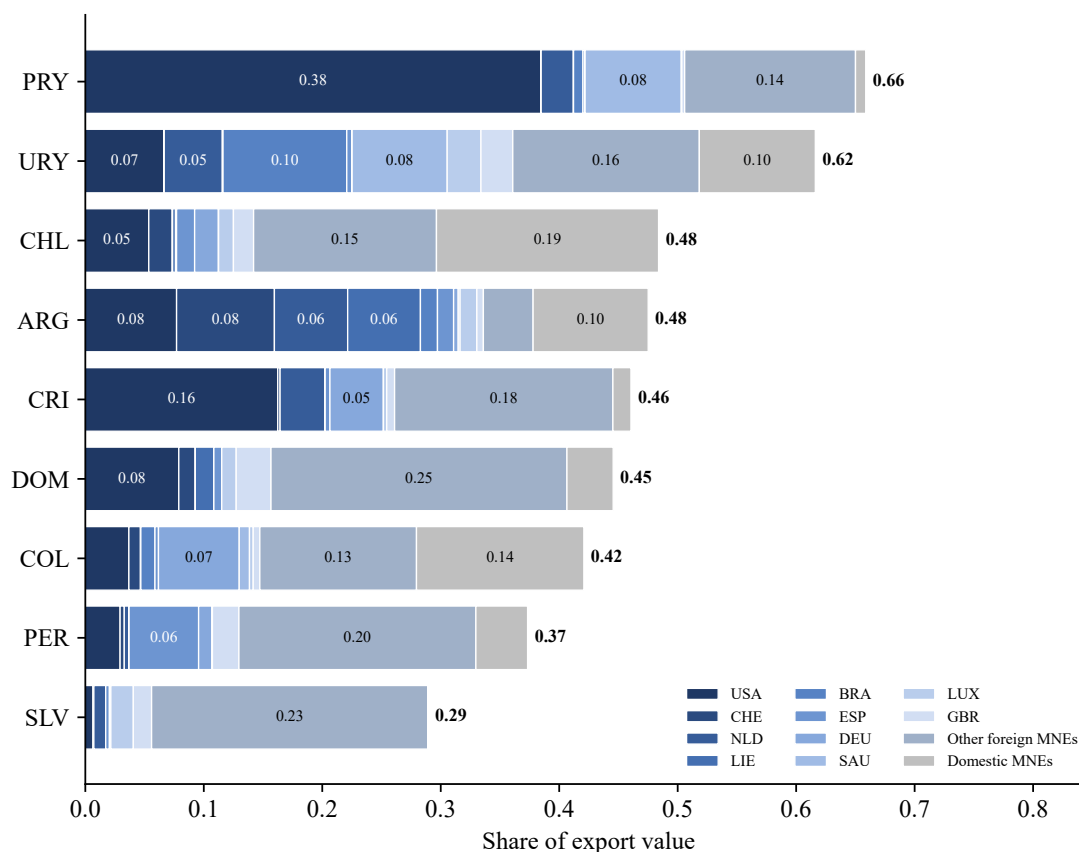


Figure A.1: Multinational Share of Agricultural Export Value, by Exporting Country and Parent Country

Notes: Shares of each country's agricultural export value. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.2: Multinational Share of Agricultural Export Value, by Exporting Country and Parent Country

Origin	USA	CHE	NLD	LIE	BRA	ESP	DEU	SAU	LUX	GBR	Other	Domestic	All MNEs	Value (US\$bn/yr)
PRY	0.385	0.000	0.027	0.000	0.008	0.002	0.000	0.081	0.000	0.002	0.144	0.009	0.659	6.2
URY	0.066	0.000	0.049	0.001	0.104	0.004	0.000	0.080	0.028	0.027	0.157	0.098	0.616	4.4
CHL	0.054	0.020	0.000	0.002	0.001	0.015	0.020	0.000	0.012	0.017	0.154	0.187	0.484	7.5
ARG	0.077	0.083	0.062	0.061	0.014	0.014	0.004	0.002	0.014	0.006	0.042	0.097	0.475	38.1
CRI	0.162	0.002	0.038	0.000	0.000	0.004	0.045	0.000	0.003	0.007	0.184	0.015	0.461	4.3
DOM	0.079	0.014	0.000	0.016	0.000	0.007	0.000	0.000	0.012	0.029	0.250	0.039	0.446	2.0
COL	0.037	0.010	0.000	0.000	0.012	0.003	0.068	0.009	0.003	0.006	0.132	0.141	0.421	7.2
PER	0.029	0.004	0.004	0.000	0.000	0.059	0.011	0.000	0.000	0.022	0.200	0.044	0.373	8.1
SLV	0.006	0.001	0.010	0.000	0.000	0.003	0.001	0.000	0.019	0.015	0.233	0.000	0.289	1.1

Notes: Shares of each country's agricultural export value. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 2

Table A.3: Multinational Share of Agricultural Export Value, by Quintile of Each Sophistication Measure

Measure	Foreign MNE share					Domestic MNE share				
	Q1	Q2	Q3	Q4	Q5	Q1	Q2	Q3	Q4	Q5
PCI	0.29	0.49	0.38	0.42	0.55	0.10	0.06	0.11	0.10	0.04
$ \sigma $ FGO (2022)	0.40	0.40	0.37	0.29	0.33	0.07	0.08	0.12	0.12	0.10
σ Broda–Weinstein	0.36	0.34	0.43	0.47	–	0.12	0.11	0.07	0.05	–
Upstreamness	0.32	0.31	0.37	0.36	0.44	0.08	0.09	0.12	0.11	0.08
Quality ladder	–	–	–	–	–	–	–	–	–	–
RHCI	0.41	0.48	0.37	0.42	0.30	0.10	0.04	0.10	0.09	0.11

Notes: Shares of export value within each quintile of HS6 products. –: fewer than 30 HS6 products carry the measure.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 3

Table A.4: Foreign-Multinational Agricultural Exports, by Parent Country

Parent country	Share (%)	Value (US\$bn/yr)
USA	23.9	6.7
CHE	11.0	3.1
NLD	9.5	2.7
LIE	7.6	2.2
BRA	3.8	1.1
ESP	3.8	1.0
DEU	3.5	0.8
SAU	3.2	0.9
LUX	2.8	0.8
GBR	2.7	0.7
JPN	2.6	0.6
PAN	2.5	0.6
FRA	2.5	0.7
CAN	1.8	0.4
BMU	1.7	0.5
Other (115 countries)	17.0	4.4
All foreign MNEs	100.0	30.7

Notes: Percent of foreign-MNE agricultural export value, by country of the ultimate parent. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.5: Destination of Foreign-Multinational Agricultural Exports, by Parent Country

Parent / Destination	USA	CHN	BRA	ARG	ESP	CHL	JPN	NLD	URY	ITA	Other destinations	Total
USA	7.3	15.5	3.7	16.7	1.2	4.1	1.8	1.7	5.5	0.7	41.8	100.0
CHE	1.7	6.7	7.2	0.2	2.6	3.4	0.4	1.2	0.7	3.7	72.3	100.0
NLD	3.3	14.4	5.1	3.9	1.4	3.6	0.6	2.8	2.4	5.6	57.0	100.0
LIE	1.3	14.4	6.1	0.0	2.9	2.6	0.4	0.9	0.2	2.2	69.1	100.0
BRA	11.4	31.2	4.9	1.9	1.2	1.7	1.1	4.3	0.9	2.2	39.2	100.0
ESP	15.5	8.3	3.0	0.4	26.1	2.5	3.9	5.8	0.2	2.9	31.4	100.0
DEU	37.3	0.6	1.8	0.4	3.5	0.9	5.0	4.3	0.7	1.8	43.6	100.0
SAU	6.4	12.3	6.9	0.1	2.3	15.8	0.1	2.6	0.3	2.1	51.1	100.0
LUX	6.5	2.5	40.8	0.0	8.1	2.0	2.1	0.3	2.8	4.5	30.4	100.0
GBR	8.9	16.9	8.5	0.6	3.5	2.7	2.2	2.9	2.1	0.8	50.9	100.0
Other parents	23.1	7.9	7.9	7.5	4.0	1.5	5.5	3.5	1.8	2.0	35.4	100.0

Notes: Percent of each parent country's agricultural exports, by destination (rows add to 100). Ten largest parent countries and ten largest destinations.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.6: Foreign-Multinational Agricultural Exports to the Parent Countries, by Parent Country

Parent / Destination	USA	CHE	NLD	BRA	ESP	DEU	SAU	LUX	GBR	Other destinations	Total
USA	7.3	0.0	1.7	3.7	1.2	0.6	1.0	0.0	0.5	84.1	100.0
CHE	1.7	0.0	1.2	7.2	2.6	0.3	3.8	0.0	0.9	82.3	100.0
NLD	3.3	0.0	2.8	5.1	1.4	0.6	1.0	0.0	0.8	85.0	100.0
LIE	1.3	0.0	0.9	6.1	2.9	0.1	3.0	0.0	0.5	85.3	100.0
BRA	11.4	0.6	4.3	4.9	1.2	5.1	0.9	0.0	1.3	70.3	100.0
ESP	15.5	0.0	5.8	3.0	26.1	1.6	0.3	0.0	1.9	45.7	100.0
DEU	37.3	0.1	4.3	1.8	3.5	7.1	0.2	0.0	1.2	44.4	100.0
SAU	6.4	0.7	2.6	6.9	2.3	1.2	2.6	0.0	1.5	75.7	100.0
LUX	6.5	0.1	0.3	40.8	8.1	0.7	0.0	0.0	0.4	43.1	100.0
GBR	8.9	0.2	2.9	8.5	3.5	2.1	0.0	0.0	1.6	72.3	100.0
Other parents	23.1	0.3	3.5	7.9	4.0	2.0	0.3	0.0	1.9	56.9	100.0

Notes: Percent of each parent country's agricultural exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.7: Destination of Multinational Agricultural Exports, by Parent Region

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	30.0	17.1	20.5	29.3	2.8	0.3	0.1	100.0
North America	47.9	8.3	10.2	25.6	7.6	0.1	0.4	100.0
Europe	27.3	10.1	22.0	31.6	8.2	0.6	0.2	100.0
Asia	14.2	29.2	24.7	29.1	2.4	0.3	0.2	100.0
Africa	29.5	15.3	30.8	13.1	9.2	0.3	1.8	100.0
Oceania	59.2	21.5	8.5	6.1	0.7	4.1	0.0	100.0
Rest of World	17.8	36.0	28.7	16.8	0.0	0.7	0.0	100.0
Domestic MNEs	20.5	16.0	20.4	34.9	7.0	1.1	0.1	100.0
Local firms	22.8	17.9	26.8	25.6	5.6	1.1	0.2	100.0

Notes: Percent of each group's agricultural exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

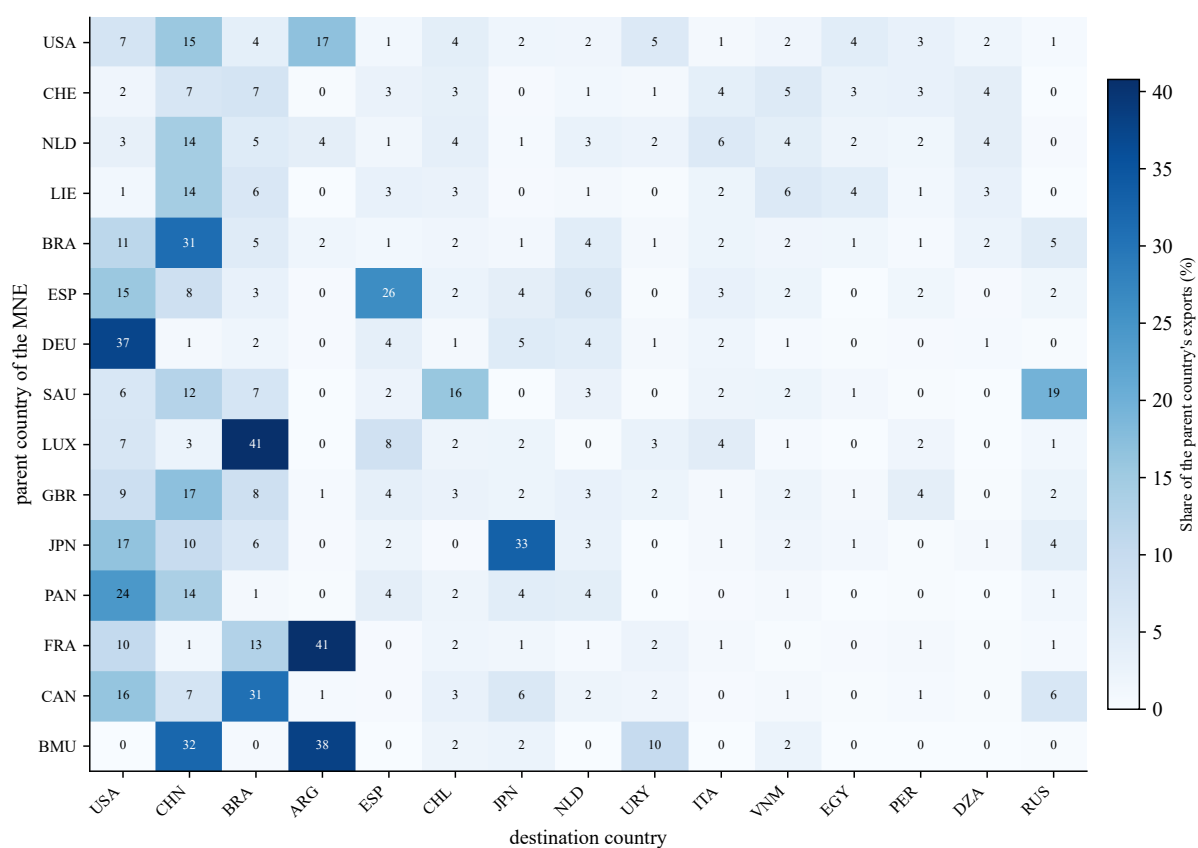


Figure A.2: Destination of Foreign-Multinational Agricultural Exports: Fifteen Largest Parent Countries and Destinations

Notes: Percent of each parent country's agricultural exports going to each destination (rows add to 100 including other destinations).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

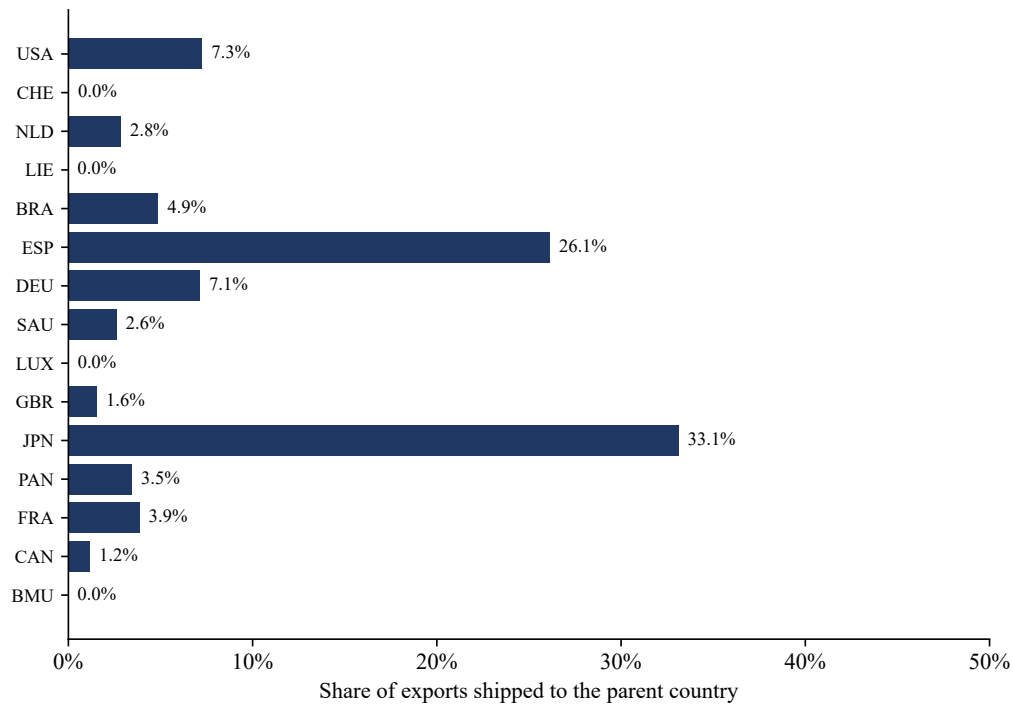


Figure A.3: Share of Foreign-Multinational Agricultural Exports Shipped to the Parent Country

Notes: Percent of each parent country's agricultural exports shipped to that same country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 4

Table A.8: Foreign-Multinational Agricultural Exports and Number of Parents, by Size of the Group's Global Network

Affiliates worldwide	Parents	Share of parents (%)	Share of foreign-MNE export value (%)
1	1,436	45.1	8.9
2–10	904	28.4	11.2
11–100	459	14.4	29.6
>100	383	12.0	50.3
All parents	3,182	100	100

Notes: Number and share of parents, and share of foreign-MNE agricultural export value, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 5

Table A.9: Multinational Presence and Agricultural Exports: Extensive Margin, by Type of Multinational

	All MNEs		Foreign / domestic		Foreign by destination, domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
Any MNE present	1.7863*** (0.0604)	0.6860*** (0.0232)				
Any foreign MNE			1.7972*** (0.0558)	0.6929*** (0.0213)		
Any foreign MNE, parent-country destination					1.6277*** (0.1122)	0.5045*** (0.0603)
Any foreign MNE, other destination					1.7886*** (0.0563)	0.6904*** (0.0217)
Any domestic MNE			1.6133*** (0.0577)	0.5683*** (0.0237)	1.5621*** (0.0556)	0.5650*** (0.0238)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
Any MNE present	1.9313*** (0.0554)	0.2812*** (0.0181)				
Any foreign MNE			1.7649*** (0.0546)	0.2854*** (0.0179)		
Any foreign MNE, parent-country destination					1.4262*** (0.1036)	0.2467*** (0.0420)
Any foreign MNE, other destination					1.7354*** (0.0537)	0.2888*** (0.0179)
Any domestic MNE			1.3082*** (0.0539)	0.2610*** (0.0201)	1.2541*** (0.0504)	0.2590*** (0.0201)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (Panel A)	295,788	268,332	295,788	268,332	295,788	268,332
Observations (Panel B)	232,617	208,479	232,617	208,479	232,617	208,479

Notes: Observations are origin–destination–HS6–year cells. Agricultural products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.10: Multinational Agricultural Exports by Exporting Country and Destination Type

Origin	MNE value (US\$bn/yr)	of which domestic (%)	Foreign-MNE value (US\$bn/yr)	% of foreign-MNE value, by destination	
				parent country	other (of which: group affiliate there)
ARG	18.1	20.4	14.4	2.8	97.2 (30.6)
CHL	3.6	38.7	2.2	12.1	87.9 (29.7)
COL	3.0	33.6	2.0	4.4	95.6 (33.7)
CRI	2.0	3.3	1.9	10.0	90.0 (45.9)
DOM	0.9	8.8	0.8	13.7	86.3 (9.7)
PER	3.0	11.7	2.7	4.7	95.3 (15.5)
PRY	4.1	1.3	4.0	1.9	98.1 (67.8)
SLV	0.3	0.0	0.3	9.5	90.5 (58.9)
URY	2.7	15.9	2.3	2.8	97.2 (42.8)
All	37.8	19.9	30.7	4.7	95.3 (35.8)

Notes: MNE and foreign-MNE agricultural export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Product groups

Table A.11: Agricultural Exports and Multinational Shares, by HS Section

HS section	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
I Live animals; animal products	12.5	6.0	34.1	14.4	SAU 0.19, JPN 0.14, ESP 0.10
II Vegetable products	33.5	16.6	41.5	7.3	USA 0.33, NLD 0.11, LIE 0.07
III Fats and oils	6.9	2.9	31.4	9.7	CHE 0.26, NLD 0.21, USA 0.19
IV Prepared foodstuffs, beverages, tobacco	25.9	12.3	37.4	9.8	USA 0.21, CHE 0.17, LIE 0.10
All	78.9	37.8	38.1	9.5	

Notes: Foreign and domestic: percent of the section's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

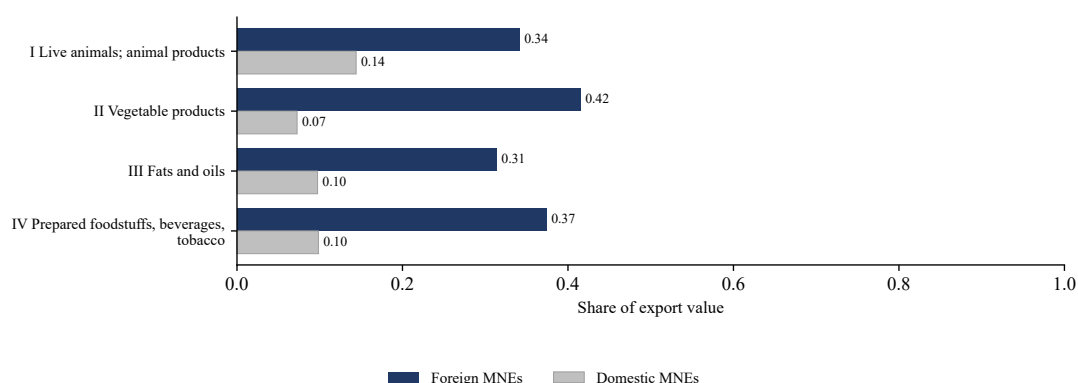


Figure A.4: Multinational Share of Agricultural Export Value, by HS Section

Notes: Shares of each section's export value, all available years pooled.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.12: Agricultural Exports and Multinational Shares, by End Use

BEC end use	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Intermediate	42.5	21.4	41.1	9.0	USA 0.30, CHE 0.16, NLD 0.13
Consumption	30.4	13.7	35.1	9.9	USA 0.16, ESP 0.09, SAU 0.08
Capital	0.0	0.0	16.8	2.0	GBR 0.35, SAU 0.33, NLD 0.28
All	72.9	35.0	38.5	9.4	

Notes: End use from BEC Rev. 4. Foreign and domestic: percent of the category's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.13: Agricultural Exports and Multinational Shares: Inputs and Other Products

Inputs vs output	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Agricultural inputs	14.9	7.0	36.0	10.8	CHE 0.26, LIE 0.18, USA 0.18
Agricultural output (food, fibres, beverages)	64.0	30.8	38.6	9.2	USA 0.25, NLD 0.09, CHE 0.08
All	78.9	37.8	38.1	9.5	

Notes: Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.14: Largest Agricultural Input Products

HS6	Description	\$bn/yr	Foreign (%)	Domestic (%)
230400	Oil-cake & other solid residues, whether/not ground/in pellets, from extraction of soyabean oil	10.84	35	12
230120	Flours, meals & pellets of fish/of crustaceans, molluscs/other aquatic invertebrates	1.70	31	10
230990	Preparations of a kind used in animal feeding other than dog/cat food put up for RS	0.51	57	3
071333	Kidney beans, incl. white pea beans (<i>Phaseolus vulgaris</i>), dried, shelled, whether/not skinned/split	0.30	28	4
230250	Bran, sharps & other residues, whether/not in the form of pellets, derived from the sifting/milling/other working of leguminous plants	0.28	30	16
100510	Maize (corn), seed	0.19	60	3
230910	Dog/cat food, put up for RS	0.17	81	5
120991	Vegetable seeds, of a kind used for sowing	0.12	67	1
230630	Oil-cake & other solid residues, whether/not ground/in pellets, from extraction of sunflower seeds	0.12	22	2
010290	Live bovine animals other than pure-bred breeding animals	0.08	31	0
071320	Chickpeas (<i>garbanzos</i>), dried, shelled, whether/not skinned/split	0.08	23	2
230800	Vegetable materials/waste/residues/by-products, whether/not in pellets, of a kind used in animal feeding, n.e.s.	0.08	19	0
100610	Rice in the husk (paddy/rough)	0.06	45	5
230110	Flours, meals & pellets of meat/meat offal; greaves	0.05	31	0
051199	Animal products not elsewhere specified/incl. (excl. of 0511.10); dead animals of Ch. 1, unfit for human consumption	0.04	39	5
071310	Peas (<i>Pisum sativum</i>), dried, shelled, whether/not skinned/split	0.03	22	1
071339	Beans (<i>Vigna</i> spp., <i>Phaseolus</i> spp. (excl. of 0713.31-0713.33)), dried, shelled, whether/not skinned/split	0.03	11	2
230230	Bran, sharps & other residues, whether/not in the form of pellets, derived from the sifting/milling/other working of wheat	0.03	16	1
010511	Live fowls of species <i>Gallus domesticus</i> , weighing not >185g	0.02	3	1
120930	Seeds of herbaceous plants cultivated principally for their flowers, of a kind used for sowing	0.02	76	2

Notes: HS6 input products with the largest export value. Foreign and domestic: percent of the product's export value. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

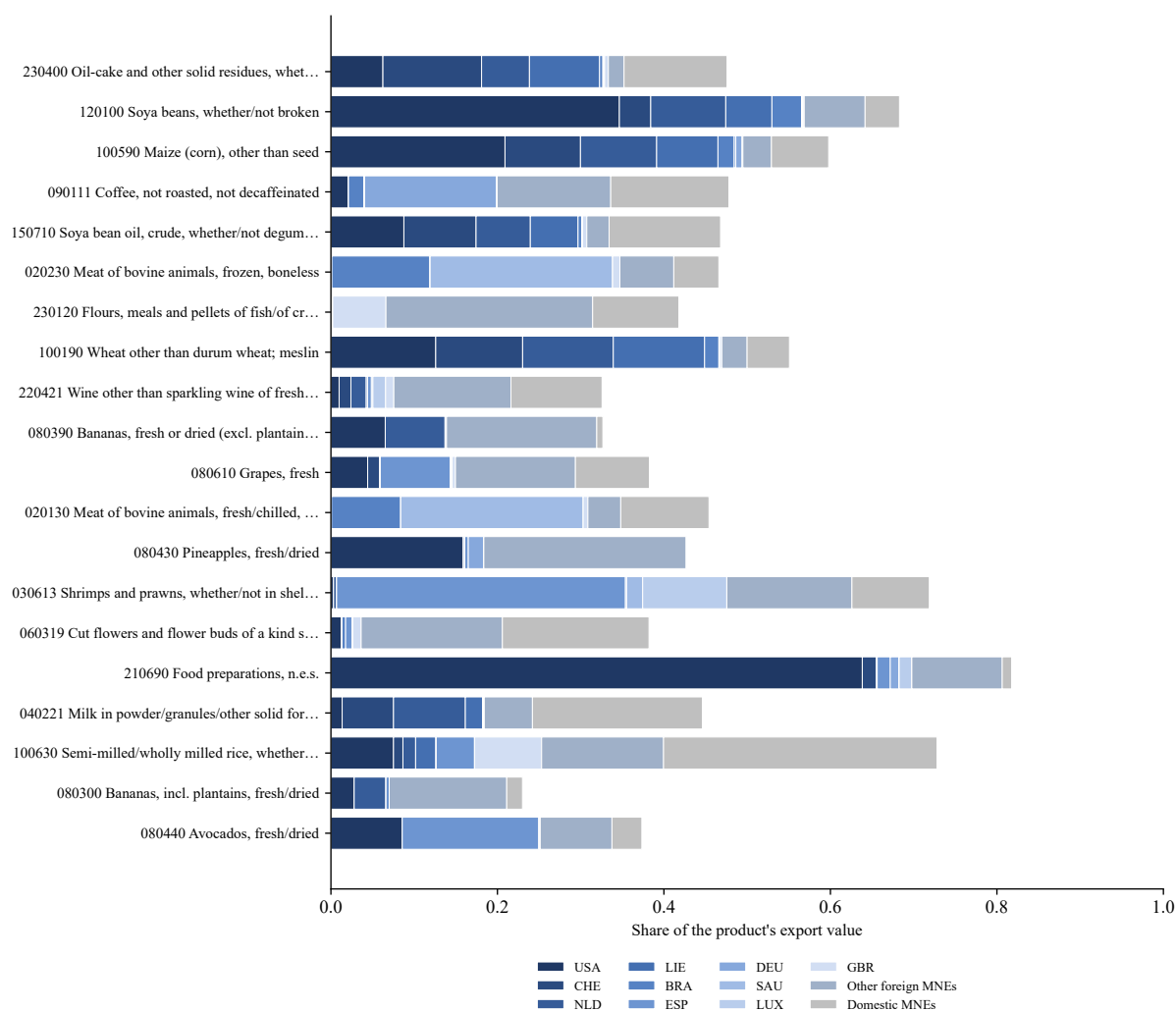


Figure A.5: Ownership of the Twenty Largest Agricultural Export Products, by Parent Country

Notes: Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

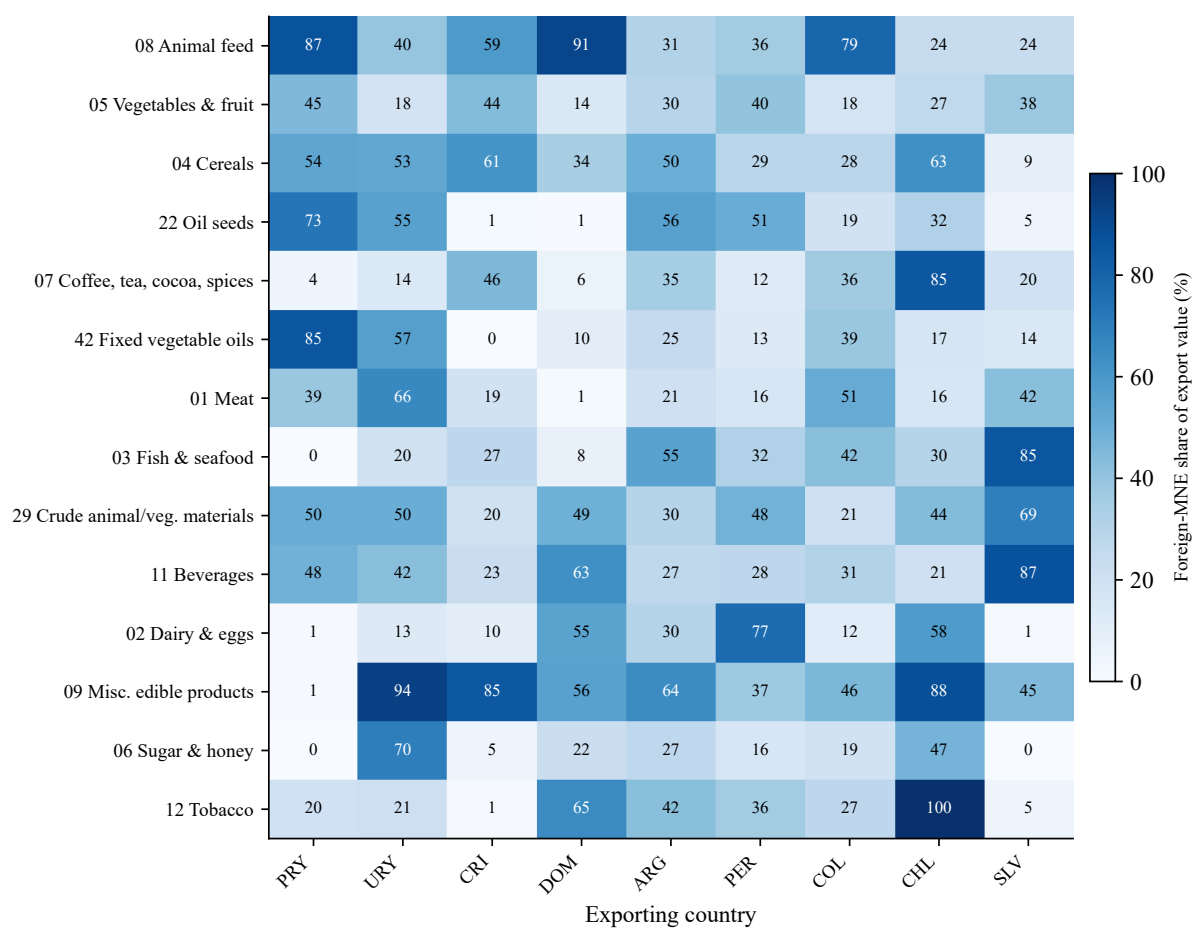


Figure A.6: Foreign-Multinational Share of Agricultural Export Value, by Exporting Country and SITC Division

Notes: Percent of each division's export value in each country shipped by foreign MNEs.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.15: Foreign-Multinational Share of Agricultural Export Value, by Exporting Country and SITC Division

Sub-sector	ARG	CHL	COL	CRI	DOM	PER	PRY	SLV	URY	All
<i>Panel A: composition of the origin's exports in the scope, % (columns sum to 100)</i>										
08 Animal feed	29.1	4.4	0.6	1.5	0.7	20.5	16.5	1.7	1.0	17.8
05 Vegetables & fruit	7.6	43.7	5.9	48.1	20.8	36.4	0.3	7.2	2.0	15.6
04 Cereals	20.9	2.8	2.5	3.6	5.6	3.3	12.5	19.6	20.5	13.4
22 Oil seeds	9.5	0.4	0.0	0.1	0.0	0.3	38.6	0.1	20.6	8.6
07 Coffee, tea, cocoa, spices	0.9	0.7	45.4	9.9	11.1	14.5	0.1	21.8	0.3	7.9
42 Fixed vegetable oils	12.8	0.5	6.2	5.2	1.2	0.5	9.1	0.2	0.1	7.7
01 Meat	5.9	9.2	1.4	2.8	0.7	0.1	18.3	1.3	30.2	7.0
03 Fish & seafood	3.9	10.4	1.6	1.8	0.6	13.1	0.0	9.6	3.0	4.9
29 Crude animal/veg. materials	0.4	3.8	23.4	3.7	1.0	2.1	1.1	1.6	1.5	3.6
11 Beverages	2.5	16.4	0.4	1.7	6.8	0.5	0.0	9.2	0.5	3.4
02 Dairy & eggs	2.8	1.4	0.3	3.9	0.6	1.5	0.2	2.4	12.7	2.6
09 Misc. edible products	1.3	3.5	2.2	15.1	6.4	0.9	0.2	5.3	2.4	2.4
06 Sugar & honey	1.1	0.5	8.8	1.1	6.7	0.8	1.1	12.1	0.6	2.1
12 Tobacco	0.9	0.8	0.6	0.3	37.1	0.1	0.8	0.0	1.1	1.5
41 Animal oils & fats	0.0	1.0	0.0	0.0	0.0	4.4	0.3	0.0	1.3	0.7
Other	0.5	0.5	0.7	0.9	0.7	1.1	0.7	7.7	2.2	0.9
All sub-sectors	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
<i>Panel B: foreign-MNE share of export value within the cell, %</i>										
08 Animal feed	30.9	24.2	79.1	58.9	91.5	35.7	86.6	24.1	39.6	35.7
05 Vegetables & fruit	29.6	26.5	17.9	44.0	14.3	40.0	45.0	38.0	18.1	32.6
04 Cereals	49.9	63.0	28.2	61.0	34.2	28.6	53.7	8.8	52.8	48.6
22 Oil seeds	55.8	31.8	18.8	1.2	0.8	50.5	73.1	4.8	55.5	61.6
07 Coffee, tea, cocoa, spices	35.4	85.1	36.2	45.6	6.3	12.4	4.0	20.2	13.7	30.4
42 Fixed vegetable oils	25.2	17.1	38.6	0.0	10.2	12.8	85.2	14.4	56.6	30.8
01 Meat	20.8	16.3	50.8	19.3	1.4	16.3	38.6	42.4	66.0	34.8
03 Fish & seafood	55.4	30.4	42.4	27.4	7.5	32.0	0.0	85.4	19.6	41.9
29 Crude animal/veg. materials	29.7	43.9	20.6	19.7	49.2	47.8	50.5	69.3	50.2	27.3
11 Beverages	27.3	20.7	31.0	22.9	63.0	27.7	48.3	87.3	42.4	28.6
02 Dairy & eggs	29.8	58.4	11.7	9.6	54.9	77.3	0.8	0.8	13.1	28.0
09 Misc. edible products	64.5	88.2	45.8	84.6	55.7	37.1	1.1	44.7	93.6	70.9
06 Sugar & honey	26.7	47.4	19.2	4.9	22.1	16.2	0.1	0.1	69.8	19.3
12 Tobacco	42.3	99.8	27.4	1.0	65.0	35.6	20.0	5.0	21.5	54.8
41 Animal oils & fats	26.3	13.3	41.6	36.8	89.4	24.9	28.7	49.8	20.5	23.0
Other	36.1	45.3	16.5	26.3	5.1	7.5	1.6	0.1	34.5	21.6
All sub-sectors	37.2	30.3	30.4	44.4	40.7	33.0	65.0	29.6	50.2	38.5

Notes: Percent of each division's export value in each country shipped by foreign MNEs.

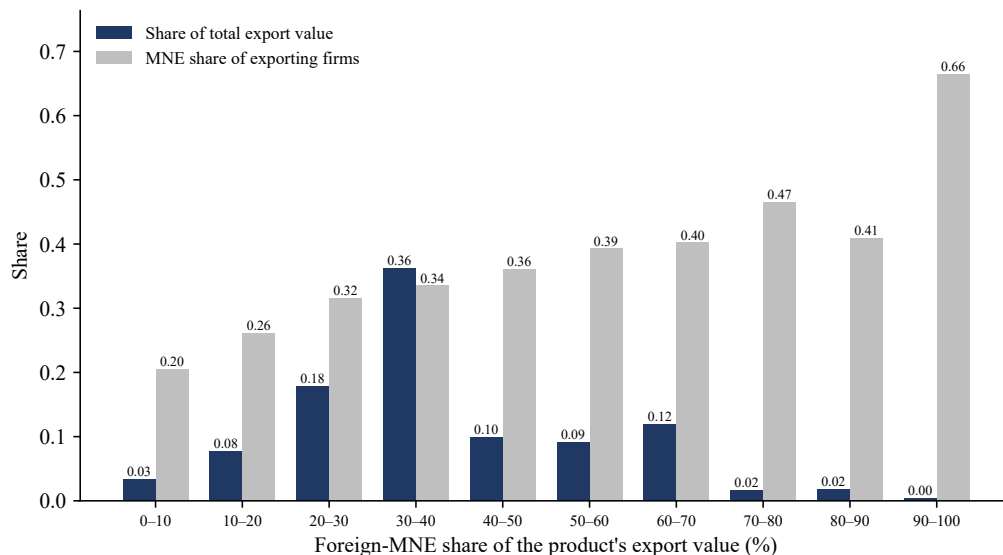
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.16: Foreign-Multinational Agricultural Exports by SITC Division and Parent Country

SITC Rev. 3 division	USA	CHE	NLD	LIE	BRA	ESP	DEU	SAU	LUX	GBR	Other	Domestic
08 Animal feed	6.3	9.8	4.7	6.5	0.3	0.5	0.2	0.1	0.0	1.2	6.2	11.5
05 Vegetables & fruit	3.2	0.2	0.7	0.0	0.0	3.3	1.2	0.1	0.8	0.4	22.7	5.8
04 Cereals	15.0	6.2	6.2	5.8	1.3	0.5	1.0	0.0	3.5	1.0	8.2	8.2
22 Oil seeds	30.7	3.2	7.8	4.8	3.1	0.1	0.1	0.0	0.0	0.3	11.5	4.1
07 Coffee, tea, cocoa, spices	1.5	0.8	0.0	0.0	1.2	0.1	10.2	0.0	1.1	0.0	15.6	15.6
42 Fixed vegetable oils	6.3	7.3	6.7	4.0	0.3	0.1	0.0	0.0	0.1	0.4	5.6	10.2
01 Meat	0.2	0.0	0.0	0.0	7.7	0.1	0.0	13.4	0.0	0.9	12.5	13.1
03 Fish & seafood	1.7	0.0	0.1	0.0	0.0	10.1	0.0	0.4	3.4	0.7	25.4	9.5
29 Crude animal/veg. materials	0.9	0.1	0.7	0.7	1.0	0.3	1.8	0.7	0.2	0.5	20.4	9.0
11 Beverages	0.7	0.6	0.8	0.1	0.0	1.2	0.1	0.0	3.6	2.6	19.0	9.2
02 Dairy & eggs	0.8	5.2	5.3	0.9	0.1	0.1	0.0	0.0	0.0	0.1	15.6	20.3
09 Misc. edible products	31.8	12.0	1.0	0.0	0.1	0.7	0.5	0.0	0.7	4.5	19.7	2.3
06 Sugar & honey	7.4	0.0	0.6	1.5	0.0	0.1	0.7	0.0	0.0	0.1	8.8	14.7
12 Tobacco	19.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.4	29.2	0.2
41 Animal oils & fats	0.0	0.0	1.7	0.0	0.4	0.0	0.0	1.4	0.0	3.3	16.2	11.0
Other	3.2	3.4	0.3	0.8	0.1	0.6	1.6	0.3	0.5	1.3	9.7	1.9

Notes: Shares of each division's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure A.7:** Distribution of Agricultural Export Value by Product-Level Foreign-Multinational Share

Notes: Share of agricultural export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.